

HENRY KOEHN

Cedar Falls, IA • 319-883-1322 • henrykoehn1@gmail.com

EDUCATION

The University of Iowa — B.S.E., Computer Science and Engineering
Iowa City, IA

Anticipated May 2027

TECHNICAL SKILLS

Languages: Java, C++, Python, SQL, JavaScript

Frameworks & Tools: Spring, Spring Boot, .NET, Blazor, Entity Framework, Docker, Jenkins, Ansible, Chef, Dynatrace

Platforms & Concepts: Azure, Linux, CI/CD, Site Reliability Engineering, Agile/Scrum, Relational Databases, Containerization

Data / ML: Pandas, NumPy, Jupyter

PROFESSIONAL EXPERIENCE

Site Reliability Engineering Intern — Mastercard

June 2026 – August 2026

O'Fallon, MO

- Co-developed a Copilot agent integrated with a Power Automate flow to automatically generate Post-Incident Reports (PIRs) from Microsoft Teams call transcripts of TRT (Technical Recovery Team) incident calls, replacing a fully manual documentation process; the solution was adopted for production implementation at Mastercard.
- Supported full Azure API Gateway outbound deployment execution across global regions (Australia, Americas, South Africa, and more) by updating service configuration versions, validating outbound endpoint mappings, and coordinating release steps across Jenkins, Ansible/XLR, and Azure gateway infrastructure.
- Coordinated with engineering teams across Pune, India and the U.S. through daily handover calls to maintain 24/7 operational continuity and align on priorities across time zones.
- Designed and built a Dynatrace dashboard providing real-time visibility into server health across the API Gateway environment, enabling faster incident detection for the SRE team.
- Automated build and configuration workflows using Jenkins, Ansible, and Chef, reducing manual steps in deployment and environment setup.
- Supported the Site Reliability Engineering (SRE) team responsible for uptime and performance of the API Gateway platform.

Research Assistant — Sinapse Lab, University of Iowa

February 2026 – Present

Iowa City, IA

- Containerized a deep learning tractography pipeline (CoRNN) using Docker, resolving GPU compatibility and multi-tool dependency conflicts across FSL, ANTs, and MRtrix3.
- Engineered and validated Docker-based builds across multiple Linux distributions to ensure reproducible, portable deployment of the research pipeline.
- Authored setup and troubleshooting documentation adopted by the lab team, streamlining onboarding and reducing environment-configuration issues.

Software Engineering Intern — Casey's

May 2025 – August 2025

Ankeny, IA

- Developed and maintained internal accounting applications using C#, .NET, SQL, Docker, Blazor, and Azure within an Agile development environment.
- Contributed to the design and implementation of a full rewrite of a Fortune 500 company's sales reconciliation application.
- Built and optimized relational database queries using Microsoft Entity Framework and SQL to manage data across multiple environments.

PUBLICATIONS

- Co-Author, "Machine Learning-Based Prediction of Infant Hypoglycemia" — manuscript in preparation with University of Iowa Stead Family Children's Hospital clinicians; planned submission to Hospital Pediatrics, September 2026.

PROJECTS

Machine Learning Pilot Research

- Partnered with clinicians at University of Iowa Stead Family Children's Hospital to develop machine learning models predicting infant hypoglycemia.
- Built end-to-end Python data preprocessing and modeling pipelines with Pandas and NumPy in Jupyter notebooks using hospital-provided clinical datasets.

Pairs Trading Backtester Application

- Built a backtesting engine in C++ and Python to evaluate statistical arbitrage (pairs trading) strategies against historical market data.